A VQE Study of the Lipkin Interaction

FYS5419 – Quantum computing and quantum machine learning

PROJECT 1

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March 18, 2026

Abstract

In this project, I study the Lipkin–Meshkov–Glick model using the Variational Quantum Eigensolver (VQE) algorithm on a sequence of increasingly complex quantum systems. Starting from single-qubit operations and Bell state entanglement, I progress through a two-level Hamiltonian with an avoided crossing, a two-qubit interacting system exhibiting interaction-driven entanglement, and finally the $J = 1$ and $J = 2$ Lipkin Hamiltonians mapped onto two and four qubits respectively. Classical diagonalization using `numpy.linalg.eigh` achieves machine-precision accuracy ($\approx 10^{-15}$) and serves as the reference throughout. Hardware-efficient VQE circuits based on R_y rotations and entangling gates reproduce exact ground state energies to $\approx 10^{-11}$ for the two-level Hamiltonian and the Lipkin $J = 1$ system. For the two-qubit Hamiltonian the VQE achieves machine precision for $\lambda \lesssim 0.4$ but undergoes a branch-tracking failure at the avoided crossing, tracking the excited state with errors of $\approx 10^{-1}$ beyond that point. For the $J = 2$ Lipkin model the residual error is $\approx 10^{-2}$, reflecting optimization in a high-dimensional symmetry-mixed landscape. Hartree-Fock results are included for comparison, and the onset of the quantum phase transition in the Lipkin model is clearly captured in both the classical and VQE spectra. All source code is available at the linked repository.

1 Introduction

Quantum computing offers a route to simulating quantum many-body systems in a way that is difficult to reproduce with classical resources, an idea often traced to early arguments by Feynman (1982) about using controllable quantum systems to emulate nature. In parallel, the modern field of quantum machine learning (QML) studies how parameterized quantum circuits can be used for inference, optimization, and simulation tasks, often within hybrid quantum–classical workflows suitable for noisy intermediate-scale quantum (NISQ) devices (Schuld and Petruccione 2018; Schuld and Killoran 2021; Conti 2024; Carleo and Troyer 2017; Carleo, Cirac, et al. 2019).

A central example of such hybrid methods is the Variational Quantum Eigensolver (VQE), which approximates the ground state of a quantum Hamiltonian using a parameterized quantum circuit evaluated on a quantum processor and optimized by a classical algorithm (Peruzzo et al. 2014; McClean, Romero, et al. 2016; Tilly et al. 2022). Variational circuit methods are closely related to other NISQ-era algorithms such as the Quantum Approximate Optimization Algorithm (QAOA) (Farhi et al. 2014; Zhou et al. 2019).

The aim of this project is to study a simplified interacting many-body Hamiltonian, known as the Lipkin model, and to compare classical diagonalization with VQE-based energy estimation across a sequence of increasingly structured test problems. I begin by constructing the computational basis for one and two qubits and studying the action of elementary operators and gates, including the Pauli matrices, Hadamard gate, phase gate, and the controlled-NOT (CNOT) gate. This provides a controlled setting for generating entangled states such as the Bell states and for quantifying bipartite entanglement through reduced density matrices and the von Neumann entropy.

I then analyze a simple two-level Hamiltonian written both as a matrix representation and in terms of Pauli operators, and I study how its eigenvalues and eigenvectors evolve as an interaction parameter is varied. This serves as a baseline for implementing the VQE algorithm and comparing its results with exact classical solutions.

Finally, I extend the analysis to interacting two-qubit systems and to the Lipkin Hamiltonian, originally introduced as a solvable model for testing many-body approximation methods (Lipkin et al. 1965). The Hamiltonian is rewritten in terms of Pauli operators so that expectation values can be evaluated using quantum circuits. Classical diagonalization results are then compared with VQE simulations, allowing us to study how interaction strength influences spectral properties, state mixing, and entanglement.

The remainder of this report is structured as follows. Section 2 introduces the quantum formalism, the VQE algorithm, and the Lipkin model. Section 3 presents results for each system in order of increasing complexity. Section 4 discusses numerical precision and the limitations of the variational approach. Section 5 summarizes the main findings and outlines directions for future work. Supplementary figures, circuit diagrams, and the Python environment are collected in the Appendix.

2 Methods

2.1 Initial definitions

First, I introduce some basic components as part of QML (Nielsen and Chuang 2000; Rieffel and Polak 2011; Olivares 2025). One-qubit bases $|0\rangle$ and $|1\rangle$ are defined in eq. (1).

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{1}$$

Using these one-qubit bases, I can write quantum states as a linear combination as shown in eq. (2), or using the Bloch-sphere representation rewritten as in eq. (3)

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle \tag{2}$$

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + \sin\left(\frac{\theta}{2}\right)e^{i\phi}\,|1\rangle \tag{3}$$

Quantum algorithms are typically formulated using Pauli operators $\sigma_x, \sigma_y, \sigma_z$ as seen in eq. (4) and a small universal set of quantum gates that act on qubits. The Pauli matrices together with the identity $\mathbf{I}$, form a convenient operator basis for expressing quantum Hamiltonians and observables.

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{4}$$

In practical implementations, these operators are realized through elementary gates such as the Hadamard gate (H), which creates superposition, the phase gate (S), which introduces controlled phase shifts, and the CNOT gate, which generates entanglement between qubits. All as seen in eq. (5). These gates form a standard set of building blocks for constructing quantum circuits and implementing algorithms such as the VQE.

$$\begin{gathered} H = \tfrac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \\ \mathrm{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \end{gathered} \tag{5}$$

2.2 Tensor products and two-qubit states

When describing systems of multiple qubits, the Hilbert space of the composite system is constructed using tensor (Kronecker) products of the individual qubit states as seen in eq. (6). This allows basis states of multi-qubit systems to be written as products of the single-qubit computational basis states $|0\rangle$ and $|1\rangle$. For a two-qubit system the computational basis therefore consists of the four states $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, obtained through tensor products of the single-qubit states.

$$\begin{aligned} |00\rangle &= |0\rangle \otimes |0\rangle \quad |01\rangle = |0\rangle \otimes |1\rangle \\ |10\rangle &= |1\rangle \otimes |0\rangle \quad |11\rangle = |1\rangle \otimes |1\rangle \end{aligned} \tag{6}$$

Certain superpositions of these two-qubit basis states form maximally entangled states known as Bell states as seen in eq. (7). These states play an important role in quantum information theory and serve as fundamental examples of quantum entanglement between two qubits (Einstein et al. 1935; Nielsen and Chuang 2000).

$$\begin{aligned} |\Phi^+\rangle &= \frac{|00\rangle + |11\rangle}{\sqrt{2}} \quad |\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}} \\ |\Psi^+\rangle &= \frac{|01\rangle + |10\rangle}{\sqrt{2}} \quad |\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}} \end{aligned} \tag{7}$$

For composite quantum systems, operators acting on multiple qubits are constructed using tensor products of single-qubit operators. If an operator A acts on the first qubit and an operator B acts on the second qubit, the combined operator acting on the two-qubit system is written as $A \otimes B$. In this way, Hamiltonians and quantum gates for multi-qubit systems can be expressed in terms of tensor products of Pauli matrices and the identity operator.

As an example, the operator $I \otimes \sigma_x$ flips the second qubit while leaving the first qubit unchanged. In the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ this operator is given in eq. (8).

$$I \otimes \sigma_x = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{8}$$

Similarly, tensor products between Pauli operators act on both qubits simultaneously. For example, the operator $\sigma_x \otimes \sigma_x$ flips both qubits as seen in eq. (9).

$$\sigma_x \otimes \sigma_x = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}. \tag{9}$$

While pure quantum states are written as state vectors, a more general description is given by density matrices. For a pure state $|\psi\rangle$ the density operator is given in eq. (10) and for a statistical ensemble $\{|\psi_i\rangle, p_i\}$ it becomes as in eq. (11).

$$\rho = |\psi\rangle\langle\psi|, \tag{10}$$

$$\rho = \sum_i p_i\,|\psi_i\rangle\langle\psi_i|. \tag{11}$$

For bipartite systems the state of a subsystem is obtained via the partial trace, in eq. (12), which is central when analysing entanglement. A standard measure of quantum uncertainty is the von Neumann entropy in eq. (13), which vanishes for pure states and is positive for mixed states.

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}), \tag{12}$$

$$\mathcal{S}(\rho) = -\mathrm{Tr}(\rho \log \rho), \tag{13}$$

2.3 Eigenvalue analysis

The stationary Schrödinger equation defines the energy spectrum of a quantum system. The Hamiltonian operator acting on a quantum state produces the same state scaled by its corresponding energy eigenvalue, as seen in eq. (14), where $\mathcal{H}$ is the Hamiltonian and E the energy eigenvalue and $|\psi\rangle$ the corresponding eigenstate.

$$\mathcal{H}\,|\psi\rangle = E\,|\psi\rangle \tag{14}$$

In the computational basis the Hamiltonian can be represented as a matrix acting on the state vector, and for a two-level system this can take the form in eq. (15). Here the diagonal elements correspond to energies of basis states while the off-diagonal elements represents interactions between them.

$$\mathcal{H} = \begin{bmatrix} \mathcal{H}_{11} & \mathcal{H}_{12} \\ \mathcal{H}_{21} & \mathcal{H}_{22} \end{bmatrix} \tag{15}$$

Furthermore, the Hamiltonian is decomposed into $\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_1$, with a solvable non-interacting component $\mathcal{H}_0$ and an interaction term $\mathcal{H}_1$. See eq. (16), where E_i represent the unperturbed energy levels and V_{ij} describe interaction matrix elements.

$$\mathcal{H}_0 = \begin{bmatrix} E_1 & 0 \\ 0 & E_2 \end{bmatrix} \quad \mathcal{H}_1 = \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix} \tag{16}$$

Any 2×2 Hamiltonian can be expressed using the Pauli operator basis together with the identity operator (Scherer 2019). For the present system this leads to the identities seen in eq. (17) with eq. (18).

$$\begin{aligned} \mathcal{H} &= a\mathbf{I} + b\sigma_x + c\sigma_y + d\sigma_z \\ \mathcal{H}_0 &= \mathcal{E}\mathbf{I} + \Omega\sigma_z \\ \mathcal{H}_1 &= c\mathbf{I} + \omega_z\sigma_z + \omega_x\sigma_x \end{aligned} \tag{17}$$

$$\mathcal{E} = \frac{E_1 + E_2}{2} \quad \Omega = \frac{E_1 - E_2}{2} \tag{18}$$

Now, I can move on to the analytical eigenvalue solution, where the eigenvalues of the Hamiltonian are obtained from the characteristic equation in eq. (19).

$$\det(\mathcal{H} - E\mathbf{I}) = 0 \tag{19}$$

For a general 2×2 Hamiltonian in eq. (20) the resulting energy eigenvalues can be found in eq. (21). Using this, the lowest eigenvalue corresponds to the ground state energy of the system.

$$\mathcal{H} = \begin{pmatrix} a & b \\ b & d \end{pmatrix} \tag{20}$$

$$E_\pm = \frac{a+d}{2} \pm \sqrt{\left(\frac{a-d}{2}\right)^2 + b^2} \tag{21}$$

To study the effects of interactions in eq. (22), the Hamiltonian is parametrised by a coupling parameter $\lambda \in [0, 1]$, where the limits $\lambda = 0$ and $\lambda = 1$ correspond to the non-interacting and fully interacting systems respectively.

$$\mathcal{H} = \mathcal{H}_0 + \lambda\mathcal{H}_1 \tag{22}$$

The Hamiltonian is diagonalised numerically to obtain its eigenvalues and eigenvectors. For Hermitian matrices this is efficiently performed using the `numpy` package in `python`, as seen in listing 1.

```
np.linalg.eigh(H)
```

Listing 1. Diagonalization of the Hamiltonian matrix $\mathcal{H}$ using NumPy's `np.linalg.eigh`, which computes the eigenvalues and eigenvectors of a Hermitian operator. This routine is used as the classical baseline for validating the quantum variational results.

2.4 Variational Quantum Eigensolver (VQE)

VQE is based on the variational principle, for any normalised trial state $|\psi(\theta)\rangle$, the expectation value of the Hamiltonian provides an upper bound to the true ground state energy. The goal in using a VQE is therefore to minimize the energy expectation value with respect to the circuit parameter θ shown in eq. (23).

$$E_0 \leq \min_\theta \langle\psi(\theta)|\,\mathcal{H}\,|\psi(\theta)\rangle \tag{23}$$

A parameterized quantum circuit $U(\theta)$ prepares a trial state from a simple initial computational basis state. By varying the parameters θ, the circuit explores different candidate states in eq. (24)

$$|\psi(\theta)\rangle = U(\theta)\,|0\rangle^{\otimes n} \tag{24}$$

To evaluate the energy on a quantum computer, the Hamiltonian is expressed as a weighted sum of tensor products of Pauli operators, in eq. (25). Each term can then be measured separately on the quantum circuit, where c_k are the real coefficients and P_k are Pauli strings acting on the qubits.

$$\mathcal{H} = \sum_k c_k P_k \quad P_k \in \{I, X, Y, Z\}^{\otimes n} \tag{25}$$

The VQE cost function is the expectation value of the Hamiltonian with respect to the trial state. Using the Pauli decomposition, the energy can be written as a sum of expectation values of Pauli operators in eq. (26).

$$E(\theta) = \langle\psi(\theta)|\,\mathcal{H}\,|\psi(\theta)\rangle = \sum_k c_k \langle P_k\rangle_\theta \tag{26}$$

A classical optimizer updates the circuit parameter iteratively in order to reduce the energy expectation value. Conceptually, this can be written as an update rule, where η is a learning rate or step size, as in eq. (27).

$$\theta_{t+1} = \theta_t - \eta\nabla_\theta E(\theta_t) \tag{27}$$

After convergence, the optimal parameters θ^* define the approximate ground state prepared by the circuit. The estimated ground state energy is then shown in eq. (28).

$$E_{\mathrm{VQE}} = E(\theta) \tag{28}$$

The choice of parameterized circuit $U(\theta)$ determines the family of states explored by the VQE and therefore strongly influences the quality of the variational solution. For a single qubit, a minimal ansatz consists of a single rotation gate is seen in eq. (29) and illustrated in fig. 1.

$$|\psi(\theta)\rangle = R_y(\theta)\,|0\rangle = \cos\frac{\theta}{2}\,|0\rangle + \sin\frac{\theta}{2}\,|1\rangle\,. \tag{29}$$

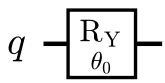

Figure 1. Minimal ansatz on a single qubit

For two qubits, a common hardware-efficient architecture combines single-qubit rotations with an entangling gate. A simple example is shown in eq. (30) and fig. 2.

$$|\psi(\boldsymbol{\theta})\rangle = \big(R_y(\theta_3)\otimes R_y(\theta_4)\big)\,\mathrm{CNOT}\,\big(R_y(\theta_1)\otimes R_y(\theta_2)\big)\,|00\rangle\,. \tag{30}$$

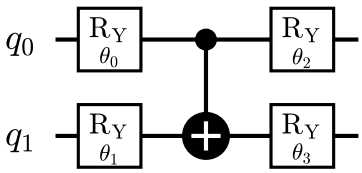

Figure 2. A simple two-qubit hardware-efficient ansatz combining layers of parameterized R_y rotations and a CNOT entangling gate, corresponding to eq. (30).

2.5 The Lipkin model

The Lipkin–Meshkov–Glick (LMG) model describes N fermions distributed over two N-fold degenerate energy levels separated by an energy ε. In terms of quasispin operators the full Hamiltonian is $\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_1 + \mathcal{H}_2$, written as in eq. (31).

$$\mathcal{H} = \varepsilon J_z + \frac{1}{2}V(J_+^2 + J_-^2) + \frac{1}{2}W(-N + J_+J_- + J_-J_+) \tag{31}$$

The operators J_z, J_+, J_- satisfy the $\mathfrak{su}(2)$ algebra and act within the angular momentum sector $J = N/2$. The first term $\mathcal{H}_0$ represents the single-particle energy splitting. The second term $\mathcal{H}_1$ scatters pairs of particles between the two levels ($\Delta m = \pm 2$). The third term $\mathcal{H}_2$, controlled by W, describes a monopole–monopole scattering that shifts states within a level without changing the level occupancy. Setting $W = 0$ recovers the simpler pair-scattering model used as the primary study case.

To simulate the model on a quantum computer, the quasispin operators are mapped to Pauli operators acting on N qubits. This yields the qubit Hamiltonian shown in eq. (32).

$$\mathcal{H} = \frac{\varepsilon}{2}\sum_{i=1}^{N} Z_i + \sum_{i<j}\left(\frac{V+W}{2}\,X_iX_j + \frac{W-V}{2}\,Y_iY_j\right) \tag{32}$$

The V and W terms both contribute to the XX and YY Pauli strings: the pair-scattering (V) drives $XX - YY$ while the monopole term (W) drives $XX + YY$, so the combined coefficients are $(V + W)/2$ and $(W - V)/2$ respectively. Setting $W = 0$ recovers the pure $XX - YY$ structure. This Pauli representation allows the full Lipkin Hamiltonian to be directly evaluated within the VQE framework.

2.6 Implementation in `python`

All simulations and numerical verification routines were implemented in `python`. Linear algebra and numerical routines were handled using `numpy` and `scipy`, data handling with `pandas`, and visualization with `matplotlib` and `seaborn`. Quantum circuits and hybrid quantum–classical optimization were implemented using `PennyLane` and `Qiskit` (NumPy Developers 2024; Xanadu AI 2024; IBM Quantum 2024). Python has become a standard platform for numerical computing and quantum algorithm prototyping due to its extensive scientific ecosystem (Pattanayak 2021; Hundt 2026; Hidary 2019).

The full implementation and supporting code are available in the project repository (Furnes 2026b; Furnes 2026a), with the development environment summarized in listing 2 in the Appendix. Parts of the implementation follow examples and structures from the quantum machine learning teaching repository by Hjorth-Jensen et al. (2026).

2.7 Use of AI-assisted development tools

AI-assisted programming tools were used during development for code generation, debugging, and documentation. These included large language models such as `ChatGPT` (OpenAI 2023), `Claude` (Anthropic 2026b), `Grok` (xAI 2026), and `Gemini` (Google DeepMind 2026), as well as coding assistants such as `GitHub Copilot`, `Antigravity` and `Cursor` (GitHub 2026; Google 2026; Anysphere 2026).

In writing this report, I downloaded `Claude Code` from Anthropic for the first time (Anthropic 2026a). This let me use an AI agent directly in my computer terminal. Moreso, I found that syncing the `Overleaf` LaTeXproject to `GitHub` and allowing `Claude Code` to access the project's Tex files, let me receive direct feedback from the AI agent.

3 Results

3.1 Single-Qubit Operations and Bell State Entanglement

Applying the Pauli operators defined in section 2 confirms the expected behaviour.

$$\sigma_x |0\rangle = |1\rangle \qquad \sigma_x |1\rangle = |0\rangle \qquad \text{bit-flip} \tag{33}$$
$$\sigma_y |0\rangle = i |1\rangle \qquad \sigma_y |1\rangle = -i |0\rangle \qquad \text{bit-flip + phase} \tag{34}$$
$$\sigma_z |0\rangle = |0\rangle \qquad \sigma_z |1\rangle = -|1\rangle \qquad \text{phase-flip} \tag{35}$$

The Hadamard gate (H) maps computational basis states to equal superpositions, $H|0\rangle = |+\rangle$ and $H|1\rangle = |-\rangle$. The phase gate introduces a relative phase of i on $|1\rangle$ while leaving $|0\rangle$ unchanged. In our results, both gates act as expected. Starting from $|00\rangle$, the circuit $H \otimes \mathbf{I}$ followed

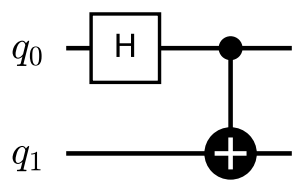

Figure 3. Quantum circuit for the preparation of the Bell state $|\Phi^+\rangle$. The two qubits are initialized in the $|00\rangle$ state. A Hadamard gate (H) is applied to the first qubit, creating a superposition state. This is followed by a CNOT gate with the first qubit as the control and the second as the target, creating the maximally entangled state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

by CNOT produced eq. (36), illustrated in fig. 3, and with repeated measurement of $|\Phi^+\rangle$ with 50 runs times 1000 shots yielded the table seen in table 1.

$$|00\rangle \xrightarrow{H\otimes\mathbf{I}} \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle \tag{36}$$

Outcome	μ	σ
$\|00\rangle$	0.4988	0.018
$\|01\rangle$	0.0000	0.000
$\|10\rangle$	0.0000	0.000
$\|11\rangle$	0.5012	0.018

Table 1. Mean measurement probabilities (μ) and standard deviations (σ) obtained from repeated circuit evaluations.

The outcomes are exclusively $|00\rangle$ and $|11\rangle$ with equal probability, which is the exact expected signature of the $|\Phi^+\rangle$ Bell state. The two qubits are perfectly correlated, measuring one immediately determines the other. The statistical fluctuations of $\sigma \approx 0.018$ are consistent with the expected noise, as in eq. (37)

$$\sqrt{\frac{p(1-p)}{N_{\text{shots}}}} \approx 0.016 \tag{37}$$

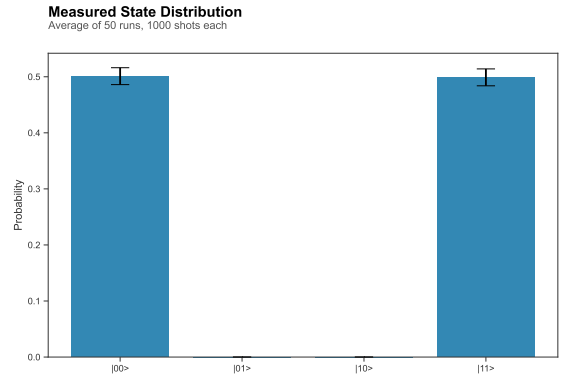

Figure 4. Measurement outcome distribution over 50 runs of 1000 shots each, showing near-equal probabilities for $|00\rangle$ and $|11\rangle$ as expected for the $|\Phi^+\rangle$ Bell state.

Next, computing the von Neumann entropy $\mathcal{S} = -\text{Tr}(\rho_A \log_2 \rho_A)$ in table 2.

State	Type	S
$\|\Phi^+\rangle$	Maximally entangled	1.000
$\|\Phi^-\rangle$	Maximally entangled	1.000
$\|\Psi^+\rangle$	Maximally entangled	1.000
$\|\Psi^-\rangle$	Maximally entangled	1.000
$\|00\rangle$	Product state	0.000
$\|+\rangle \otimes \|0\rangle$	Product (superposition)	0.000

Table 2. Von Neumann entropy $\mathcal{S}$ for several two-qubit states. The Bell states $|\Phi^\pm\rangle$ and $|\Psi^\pm\rangle$ are maximally entangled with $\mathcal{S} = 1$, while product states have vanishing entropy.

The entropy $\mathcal{S} = 1$ bit for all four Bell states confirms they are maximally entangled: no local description of either qubit alone can capture the full state. Measuring one qubit instantly fixes the outcome of the other, regardless of the basis. For the product states $|00\rangle$ and $|+\rangle\otimes|0\rangle$, the entropy vanishes exactly. The superposition state $|+\rangle \otimes |0\rangle$ is instructive: even though qubit A is in a superposition, the two qubits carry no correlations with each other, and the reduced density matrix of either subsystem is already pure. This underscores that von Neumann entropy measures entanglement, not superposition or classical uncertainty.

3.2 Two-Level System: Exact Diagonalization vs. VQE

Using parameters $E_1 = 0, E_2 = 4, V_{11} = 3, V_{22} = -3, V_{12} = V_{21} = 0.2$ the Pauli decompositions are shown in table 3.

The eigenvalues $E_\pm(\lambda)$ are computed both analytically and numerically (using `numpy.linalg.eigh`) over 101 values of $\lambda \in [0, 1]$, where the maximum discrepancy between the two methods are $\approx 10^{-15}$, as shown in fig. 5.

The spectrum shows an avoided crossing at $\lambda = \frac{2}{3}$, where at this point, the diagonal matrix elements cross ($3\lambda = 4 - 3\lambda$), but the off-diagonal coupling $V_{12} = 0.2$

Coefficient	Value	Expression
$\mathcal{E}$	2.0	$\frac{E_1+E_2}{2}$
Ω	-2.0	$\frac{E_1-E_2}{2}$
c	0.0	$\frac{V_{11}+V_{22}}{2}$
ω_z	3.0	$\frac{V_{11}-V_{22}}{2}$
ω_x	0.2	V_{12}

Table 3. Pauli decomposition coefficients from the project

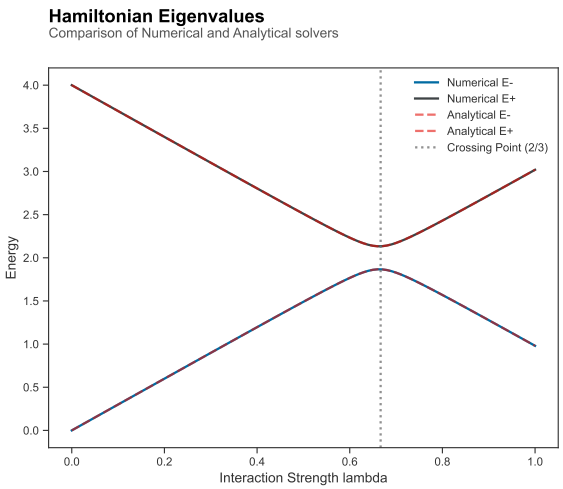

Figure 5. Eigenvalues $E_\pm(\lambda)$ as a function of the interaction parameter λ, obtained by both analytical and numerical diagonalization. An avoided crossing occurs at $\lambda = \frac{2}{3}$.

prevents a true degeneracy, creating a gap of $\Delta E \approx 0.27$.

Tracking the eigenvector decomposition of the ground state as a function of λ, as seen in table 4.

λ	Prob($\|0\rangle$)	Prob($\|1\rangle$)	Character
0.0	100%	0%	Pure $\|0\rangle$
$\frac{2}{3}$	$\sim$50%	$\sim$50%	Maximally mixed
1.0	$\sim$1%	$\sim$99%	Dominated by $\|1\rangle$

Table 4. Measurement probabilities of the computational basis states $|0\rangle$ and $|1\rangle$ as a function of the interaction parameter λ.

The ground state character inverts as λ passes through the crossing point. Below $\lambda = \frac{2}{3}$ it is predominantly $|0\rangle$, and above, $|1\rangle$ dominates. This transition is smooth because the coupling V_{12} mixes the two basis states. At $\lambda = 1$ only $\approx 1\%$ of $|0\rangle$ remains, which is consistent with the theoretical prediction.

The trial state is shown in eq. (38), where the single-parameter rotation spans the full real single-qubit Hilbert space, since the Hamiltonian is real and symmetric, R_y is exactly sufficient.

$$|\psi(\theta)\rangle = R_y(\theta)\,|0\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + \sin\left(\frac{\theta}{2}\right)|1\rangle \tag{38}$$

The energy is evaluated as $E(\theta) = c_0 + c_z\langle\sigma_z\rangle_\theta + c_x\langle\sigma_x\rangle_\theta$ where $\langle\sigma_z\rangle$ is obtained from computational-basis measurement and $\langle\sigma_x\rangle$ requires a Hadamard rotation before

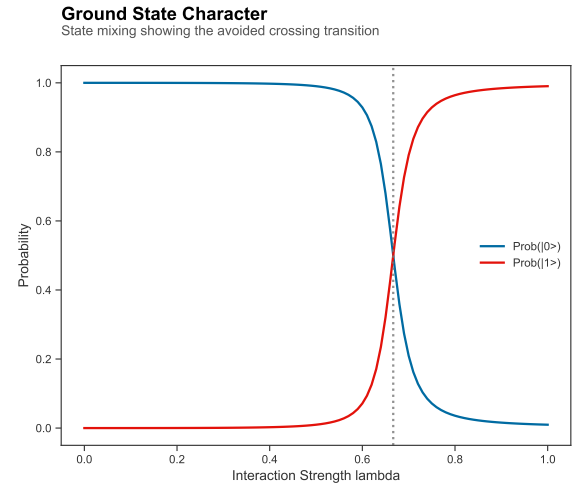

Figure 6. Ground state character as a function of λ, showing the smooth transition from a $|0\rangle$-dominated state to a $|1\rangle$-dominated state through the avoided crossing.

measurement.

The VQE is swept over $\lambda \in [0, 1]$. Both the ground state and the excited state accessed via $\theta_{\text{opt}} + \pi$ are computed. The VQE markers lie exactly on the exact diagonalization curves for both eigenvalues. Precision is shown in fig. 8 where the max error for the ground state is 5.41×10^{-11} and excited state is same order.

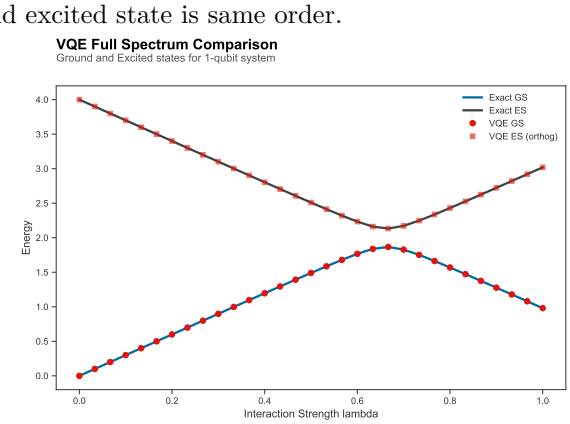

Figure 7. Full VQE spectrum showing both the ground state and excited state energies as a function of λ. VQE markers lie exactly on the exact diagonalization curves.

As the error remains below 10^{-10} across the entire λ range, this confirms that (i) the R_y ansatz is exact for this 1-qubit problem, (ii) the BFGS optimizer converges to the global minimum reliably, and (iii) the one-parameter landscape has no local minima.

An independent VQE-implementation using `Qiskit`'s `SparsePauliOp` and `Statevector` produces identical results, validating the manual implementation.

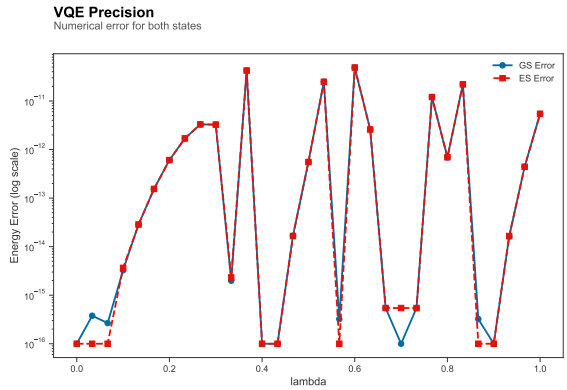

Figure 8. Absolute error between the VQE energy and exact diagonalization as a function of λ, remaining below 10^{-10} across the full range for both ground and excited states.

3.3 Two-Qubit System and Interaction-Driven Entanglement

The two-qubit Hamiltonian with parameters $H_x = 2.0, H_z = 3.0, \epsilon_{00} = 0.0, \epsilon_{10} = 2.5, \epsilon_{01} = 6.5, \epsilon_{11} = 7.0$ is diagonalized over $\lambda \in [0, 1]$.

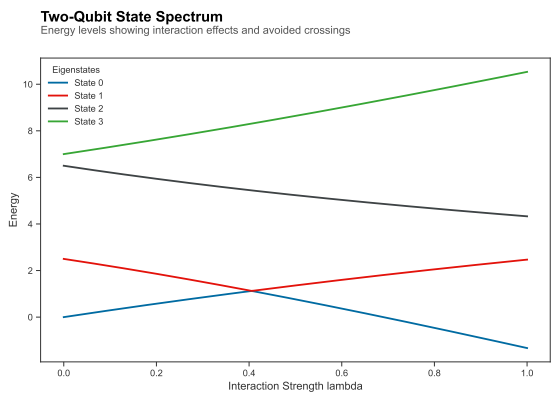

Figure 9. Eigenvalues of the two-qubit Hamiltonian as a function of λ, showing pairwise avoided crossings driven by the $\sigma_x \otimes \sigma_x$ coupling.

At $\lambda = 0$ the eigenvalues are the non-interacting energies $0.0, 2.5, 6.5, 7.0$. As λ increases, the $\sigma_z \otimes \sigma_z$ interaction shifts the diagonal elements, pushing $|00\rangle$ and $|11\rangle$ up while pulling $|10\rangle$ and $|01\rangle$ down. The $\sigma_x \otimes \sigma_x$ coupling then creates off-diagonal mixing between $|00\rangle \leftrightarrow |11\rangle$ and $|01\rangle \leftrightarrow |10\rangle$, producing avoided crossings. The block structure of the Hamiltonian, two independent 2×2 blocks, is reflected in the pairwise behaviour of the energy levels.

The von Neumann entropy is computed for all four eigenstates, not just the ground state.

At $\lambda = 0$ all eigenstates coincide with the computational basis states and carry no bipartite correlations, so $\mathcal{S} = 0$ exactly. As λ grows, the off-diagonal $\sigma_x \otimes \sigma_x$ coupling mixes the basis states within each 2×2 block, generating entanglement that increases smoothly with interaction strength.

The most striking feature is the sharp entropy spike near

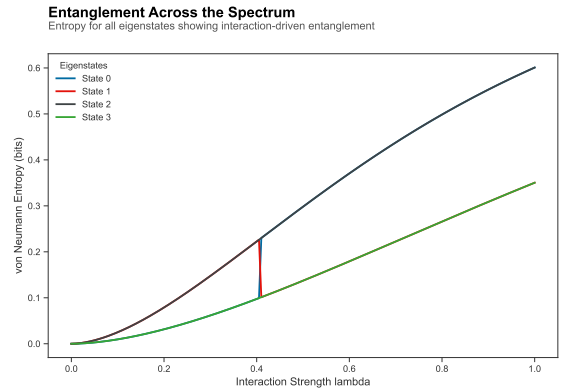

Figure 10. Von Neumann entropy $\mathcal{S}$ for all four eigenstates as a function of λ, showing monotonic growth of entanglement with a sharp rise near the avoided crossing.

the avoided crossings. At these points the eigenstate character undergoes a rapid transition — the ground state shifts from being dominated by a single product state to being a strongly entangled superposition of two. A small change in λ near the crossing therefore produces a disproportionate change in quantum correlations. This is not a numerical artifact; it is a direct consequence of the near-degeneracy at the level repulsion, where the two hybridizing states contribute roughly equally to the eigenstate and maximise the bipartite entropy.

The entanglement is entirely Hamiltonian-driven. It is not imposed externally but emerges from the competition between the diagonal energy spacings ϵ_{ij} and the off-diagonal coupling H_x. When λH_x becomes comparable to the gap between two levels, the corresponding eigenstates can no longer be written as product states and the von Neumann entropy rises sharply. The entropy is therefore a sensitive diagnostic of the balance between single-particle energies and interaction strength, and its peak coincides precisely with the strongest level mixing seen in fig. 9.

The non-interacting Hamiltonian $\mathcal{H}_0 = \mathrm{diag}\,(\epsilon_{00}, \epsilon_{01}, \epsilon_{10}, \epsilon_{11})$ is decomposed into four Pauli strings, as seen in eq. (39), with $h_0 = 4.0, h_1 = -0.75, h_2 = -2.75, h_3 = -0.5$. The interaction adds $(h_3 + \lambda\mathcal{H}_z), ZZ + \lambda\mathcal{H}_x, XX$, giving 5 Pauli terms total.

$$\mathcal{H}_0 = h_0\, II + h_1\, ZI + h_2\, IZ + h_3\, ZZ \tag{39}$$

The hardware-efficient ansatz consists of two R_y rotation layers separated by a CNOT entangler shown in eq. (40) and fig. 11.

$$|\psi(\theta)\rangle = (R_y(\theta_3) \otimes R_y(\theta_4)) \cdot \mathrm{CNOT} \cdot (R_y(\theta_1) \otimes R_y(\theta_2))\, |00\rangle \tag{40}$$

This gives 4 variational parameters. The R_y rotations span the real subspace of each qubit, and the CNOT provides the entanglement needed to represent correlated states. For a 4-dimensional real Hilbert space, 4 real parameters are sufficient.

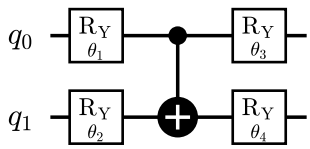

Figure 11. Quantum circuit for the two-qubit variational ansatz. The state preparation begins with the qubits in the $|00\rangle$ state. An initial layer of parameterized single-qubit Y-rotations ($R_y(\theta_1)$ and $R_y(\theta_2)$) is applied, followed by an entangling CNOT gate. A final layer of single-qubit Y-rotations ($R_y(\theta_3)$ and $R_y(\theta_4)$) completes the circuit, generating the state $|\psi(\theta)\rangle$

The VQE sweep over $\lambda \in [0, 1]$ compares three approaches in fig. 12. For $\lambda \lesssim 0.4$ all three curves are indistinguishable, with errors at machine precision as shown in fig. 13. At $\lambda \approx 0.4$ the exact ground state energy reaches a peak of ≈ 1.1 and then drops sharply, falling to ≈ -1.3 at $\lambda = 1$. Both VQE implementations, however, continue upward past the crossing, reaching ≈ 2.5 at $\lambda = 1$ — the same trajectory as the first excited state.

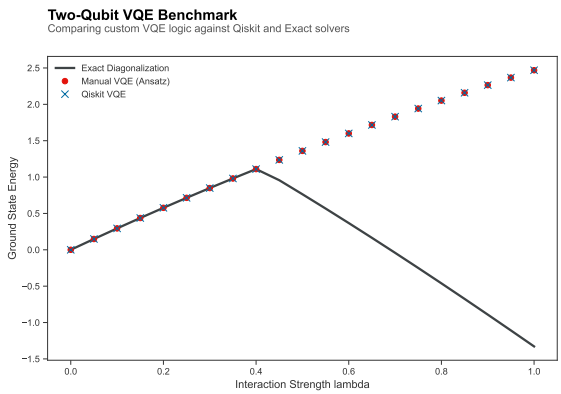

Figure 12. Ground state energy as a function of λ for three approaches: exact diagonalization, manual VQE, and Qiskit VQE. All three agree for $\lambda \lesssim 0.4$. After the avoided crossing at $\lambda \approx 0.4$ the exact ground state drops sharply while both VQE methods continue along the upper (excited-state) branch.

The failure is a branch-tracking problem. The R_y-CNOT-R_y ansatz is in principle exactly expressive for any real two-qubit state — the ansatz itself is not the limitation. The issue is that BFGS, initialized from $\boldsymbol{\theta} = \mathbf{0}$ at each λ value, follows the energy landscape continuously. Before the crossing it finds the true ground state reliably. At the crossing the optimizer has no mechanism to detect that the lower branch has switched character; it stays on the branch it has been tracking, which becomes the first excited state for $\lambda > 0.4$.

The fact that both the manual and Qiskit implementations produce identical errors throughout — including in the failure regime — is itself a meaningful result. It rules out a code defect in either implementation and confirms that the breakdown is a systematic consequence of the optimization strategy, not an artifact. Both implementations correctly handle the ansatz, Pauli measurements, and optimization

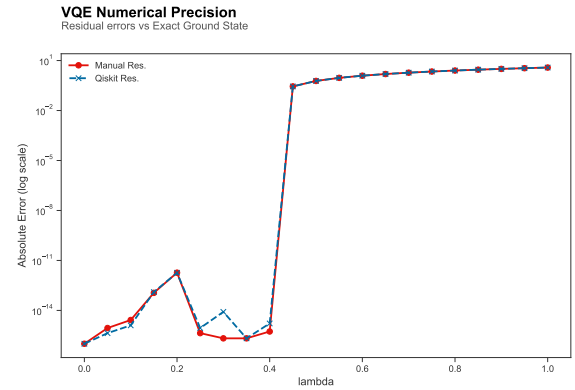

Figure 13. Absolute error of both VQE implementations relative to exact diagonalization. Below $\lambda \approx 0.4$ errors remain at machine precision (10^{-14}–10^{-11}). At the crossing the error jumps by twelve orders of magnitude and plateaus at $\approx 10^{-1}$ for all $\lambda > 0.4$, with both implementations failing identically.

loop; they simply make the same physically motivated mistake at the same crossing point.

For $\lambda \lesssim 0.4$, where the VQE does track the ground state, the entanglement structure is correctly captured: as λ increases toward the crossing the optimal rotation angles shift so that the CNOT produces increasing entanglement, consistent with the entropy growth in fig. 10. This correspondence breaks down past the crossing, where the VQE state corresponds to the first excited state rather than the ground state.

3.4 The Lipkin Model

The full Lipkin Hamiltonian including both the pair-scattering term (V) and the monopole term (W) is constructed from the quasispin operators J_z, J_+, J_-. The resulting matrices are shown for $J = 1$ with $N = 2$ particles (3×3) in eq. (41) and $J = 2$ with $N = 4$ particles (5×5) in eq. (42).

The pair-scattering term V only connects $m \leftrightarrow m \pm 2$ states (the $\Delta m = \pm 2$ selection rule from $J_+^2 + J_-^2$), so the $m = 0$ state is decoupled from the V interaction for all V. The W term, however, contributes a diagonal shift: for $J = 1$ it raises the $m = 0$ energy by W, and for $J = 2$ it shifts the odd-m states by $3W$ and the $m = 0$ state by $4W$, as visible in the matrices below.

$$H_{J=1} = \begin{pmatrix} -\epsilon & 0 & -V \\ 0 & W & 0 \\ -V & 0 & \epsilon \end{pmatrix} \tag{41}$$

The chequerboard pattern $\Delta m = \pm 2$ persists, decoupling even-m and odd-m states into separate blocks.

$$H_{J=2} = \begin{pmatrix} -2\epsilon & 0 & \sqrt{6}V & 0 & 0 \\ 0 & -\epsilon+3W & 0 & 3V & 0 \\ \sqrt{6}V & 0 & 4W & 0 & \sqrt{6}V \\ 0 & 3V & 0 & \epsilon+3W & 0 \\ 0 & 0 & \sqrt{6}V & 0 & 2\epsilon \end{pmatrix} \tag{42}$$

Mapping the quasispin operators to $N = 2$ qubits gives the full two-qubit Hamiltonian

$$H_{J=1} = \frac{\epsilon}{2}(ZI + IZ) + \frac{V+W}{2}XX + \frac{W-V}{2}YY \tag{43}$$

The 4×4 Pauli representation contains both the $J = 1$ triplet ($d = 3$) and the $J = 0$ singlet ($d = 1$). Projecting onto the triplet subspace, spanned by $|11\rangle$ ($m = -1$), $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ ($m = 0$), and $|00\rangle$ ($m = +1$), exactly reproduces the 3×3 quasispin matrix in eq. (41), including the W shift on the $m = 0$ diagonal.

For $J = 2$, four-qubit mapping:

$$\boxed{H_{J=2} = \frac{\varepsilon}{2}\sum_{i=1}^{4} Z_i + \sum_{i<j}\left(\frac{V+W}{2}X_iX_j + \frac{W-V}{2}Y_iY_j\right)}$$

This gives 4 single-qubit Z terms and 12 two-qubit interaction terms (XX and YY for each of the 6 pairs), totalling 16 Pauli strings regardless of W. The W term modifies the coefficients of the existing strings rather than introducing new ones. The 16×16 Hilbert space contains $J = 0, 1, 2$ sectors; the five $J = 2$ eigenvalues were extracted from the full Pauli spectrum and verified against the quasispin reference at a test point ($\epsilon = 1$, $V = 0.5$, $W = 0.2$).

The primary V-sweeps use $W = 0$ as the baseline, and the Pauli decomposition with $W \neq 0$ was verified to match the quasispin Hamiltonian exactly for both $J = 1$ and $J = 2$. Setting $W = 0$ recovers the pure pair-scattering model; including W breaks the residual symmetry between the XX and YY channels and lifts the $m = 0$ degeneracy seen at $W = 0$.

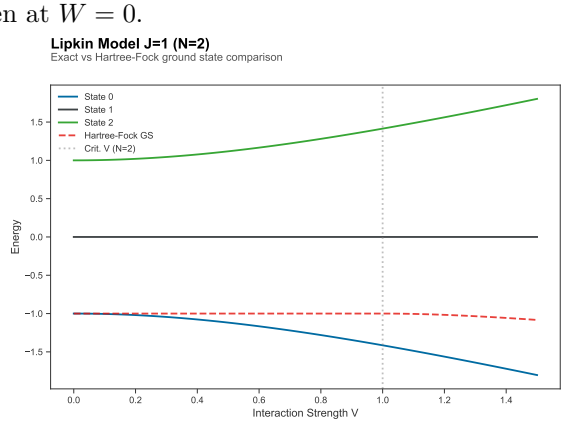

Figure 14. Lipkin model eigenvalues for $J = 1$ ($N = 2$) as a function of V, with $\epsilon = 1$ and $W = 0$.

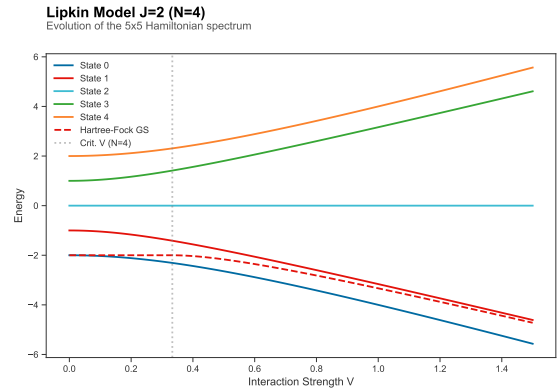

Figure 15. Lipkin model eigenvalues for $J = 2$ ($N = 4$) as a function of V, with $\epsilon = 1$ and $W = 0$.

Both spectra show the ground state energy decreasing with increasing V, reflecting energetically favorable pair correlations introduced by the interaction.

The Hartree-Fock energy of the Lipkin model is given in eq. (44), where $\tilde{V} = \frac{(N-1)V}{\epsilon}$, and the critical interaction strengths are $V_c = 1.000$ for $N = 2$ ($J = 1$) and $V_c = 0.333$ for $N = 4$ ($J = 2$).

$$E_{\text{HF}} = \begin{cases} \frac{-N\epsilon}{2} & \text{if } \tilde{V} \leq 1 \\ \frac{-N\epsilon}{4}\left(\tilde{V} + \frac{1}{\tilde{V}}\right) & \text{if } \tilde{V} > 1 \end{cases} \tag{44}$$

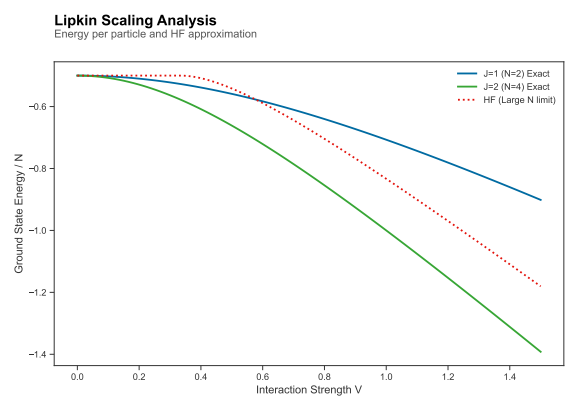

Figure 16. Ground state energy per particle E_0/N compared with the Hartree-Fock prediction for increasing system sizes, showing convergence of HF to the exact result in the thermodynamic limit.

The energy per particle $\frac{E_0}{N}$ approaches the Hartree-Fock curve as N increases, consistent with HF becoming exact in the thermodynamic limit. Below V_c HF gives the exact answer, the ground state is a single Slater determinant. Above V_c HF systematically overestimates the energy, the correlation energy is missing.

A hardware-efficient ansatz with R_y rotations and CZ entanglement is used and shown in table 5.

The ansatz uses alternating CZ entanglement layers: even layers entangle pairs $(0, 1)$ and $(2, 3)$, while odd layers entangle $(1, 2)$ and $(3, 4)$ plus a wrap-around $(0, 3)$ link to ensure full qubit connectivity after two layers. Multiple

Sys	Qbts	Depth	Params	Restarts	Opt
$J = 1$	2	2	4	3	L-BFGS-B
$J = 2$	4	4	16	4	L-BFGS-B

Table 5. Variational quantum circuit configurations used for the Lipkin model simulations.

random restarts mitigate local minima. With this I find a

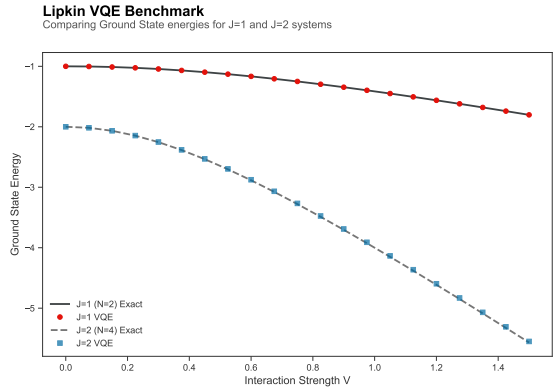

Figure 17. VQE ground state energies for both $J = 1$ and $J = 2$ Lipkin systems compared with exact diagonalization over $V \in [0, 1.5]$.

maximum error of 2.92×10^{-11} and 1.80×10^{-2} for $J = 1$ ($N = 2$) and $J = 2$ ($N = 4$) respectively, with mean errors of $\approx 10^{-12}$ and $\approx 10^{-3}$.

For the $J = 1$ system, the two-qubit R_y-CZ-R_y ansatz achieves near-exact accuracy. The low dimensionality of the problem and the block structure of the Lipkin Hamiltonian mean the variational landscape is smooth, and BFGS converges reliably to the global minimum. For the $J = 2$

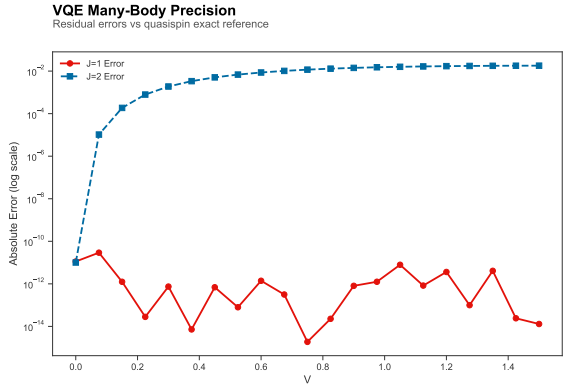

Figure 18. Absolute VQE error for the Lipkin model. The $J = 1$ system achieves machine-precision accuracy ($\approx 10^{-11}$), while the $J = 2$ system shows a larger residual error ($\approx 10^{-2}$) due to the higher-dimensional optimization landscape.

system the residual error of $\approx 10^{-2}$ is substantially larger and reflects several compounding difficulties. First, the full 16×16 qubit Hilbert space contains three symmetry sectors ($J = 0, 1, 2$); the hardware-efficient ansatz imposes no symmetry constraint, so the optimizer must implicitly discover the correct sector from the energy landscape alone. This increases the effective search space and makes convergence to the true $J = 2$ ground state less reliable. In practice, the optimized state may be a mixture of sectors or a local minimum in the $J = 0$ or $J = 1$ region that happens to have a lower apparent energy than the true $J = 2$ ground state for some parameter values.

Second, the error is not uniform across the V range: it grows near the phase transition at $V_c \approx 0.333$, where the ground state character changes rapidly and the energy landscape becomes flat around the crossing point. Near-flat landscapes cause gradient-based optimizers to stall, leading to poor convergence in exactly the regime of greatest physical interest.

Third, with 16 parameters and a circuit of depth 4, the optimization landscape exhibits the signatures of the barren plateau problem (McClean, Boixo, et al. 2018) — gradients of the cost function become exponentially small in regions of parameter space that are far from the solution, making it difficult for gradient-based methods to find a productive direction. The four random restarts partially compensate for this, but they are not sufficient to guarantee global convergence in all cases.

Both systems capture the qualitative behavior of the phase transition: the ground state energy decreases with increasing V, and the transition from weak to strong coupling is reflected in a change in the rate of energy decrease. The Hartree-Fock approximation correctly predicts the ground state below the critical coupling but underestimates correlation energy above it, consistent with the scaling analysis in fig. 16.

The VQE framework scales in qubit count as N for the Lipkin model, but the circuit depth and number of Pauli terms grow as $\mathcal{O}(N^2)$, making it increasingly challenging to maintain precision for large N on near-term hardware.

4 Numerical Precision and Limitations

Classical diagonalization caused no numerical difficulties throughout. The Hamiltonians involved are small — the largest is 5×5 — and `numpy.linalg.eigh` reproduces analytical eigenvalues to within $\approx 10^{-15}$ across all parameter ranges, including near avoided crossings where the energy gap narrows. No ill-conditioning was detected at any of the interaction strengths studied.

The VQE experiments are more varied. All energy evaluations use exact statevector simulation, so finite-shot noise plays no role in the optimization. The only exception is the Bell state measurement, which used 1000 shots per run; the observed spread ($\sigma \approx 0.018$) is consistent with binomial sampling noise at that shot count. For all other VQE runs the dominant error source is optimizer convergence.

For the two-level Hamiltonian the single-parameter R_y ansatz is exactly expressive, and residual errors of $\approx 10^{-11}$ across the full λ range reflect only the BFGS convergence tolerance. The landscape has a single minimum and a

cold start from $\theta = 0$ is sufficient throughout.

The two-qubit case is more nuanced. The R_y-CNOT-R_y ansatz is equally expressive — it can represent any real two-qubit state — and for $\lambda \lesssim 0.4$ it achieves machine-precision errors (10^{-14}–10^{-11}). However, at the avoided crossing near $\lambda \approx 0.4$ the optimizer fails to follow the ground state to the lower branch and instead tracks the first excited state for all $\lambda > 0.4$, producing errors of $\approx 10^{-1}$. This is not a local-minimum problem in the conventional sense; the optimizer converges correctly at each λ, but to the wrong eigenstate. The failure is a branch-tracking limitation of independent cold-start optimization: without an explicit strategy for crossing between branches, the optimizer remains on the continuously connected upper branch.

The Lipkin $J = 2$ system is where precision genuinely degrades. Mapping four particles to qubits embeds the problem inside a 16×16 Hilbert space that also contains $J = 0$ and $J = 1$ sectors; since the hardware-efficient ansatz carries no symmetry constraint, the optimizer must locate the physical ground state within the full unconstrained space. This is manageable away from the phase transition, but near $V_c \approx 0.333$ the ground state changes character rapidly and the cost-function gradient becomes small, causing L-BFGS-B to stall. The resulting error of $\approx 10^{-2}$ — roughly six orders of magnitude worse than the $J = 1$ case — is not a generic optimizer failure; it reflects the structural mismatch between the Hamiltonian's symmetry and the unconstrained circuit. Multiple random restarts provide only partial relief. A symmetry-adapted ansatz confined to the $J = 2$ subspace would address this directly, and is the most obvious improvement to pursue.

All results reported here assume exact statevector simulation. On actual NISQ hardware, gate errors, decoherence, and readout noise would add to the optimization difficulty already present in the $J = 2$ case. The precision figures throughout this report should therefore be read as an optimistic upper bound, not as predictions of device-level performance.

5 Conclusion

5.1 Summary and Discussion

This project worked through a sequence of increasingly complex quantum systems: from single-qubit operations and Bell state entanglement, through two-level and two-qubit Hamiltonians, to the Lipkin–Meshkov–Glick model for $N = 2$ and $N = 4$ particles. At each stage, classical diagonalization provided exact reference solutions against which the VQE results were compared.

For the two-level Hamiltonian the variational approach performs essentially perfectly: the single-parameter R_y ansatz is exactly expressive, and errors below 10^{-10} across the full λ range confirm the ansatz is not the limiting factor. The two-qubit case is more revealing. The R_y-CNOT-R_y circuit is also exactly expressive, and for $\lambda \lesssim 0.4$ it matches the exact ground state to machine precision. At the avoided crossing near $\lambda \approx 0.4$, however, both VQE implementations diverge from the exact solution: the optimizer fails to follow the ground state to the lower branch and instead tracks the first excited state, with errors jumping from $\sim 10^{-11}$ to $\sim 10^{-1}$ at the crossing. This is a branch-tracking failure rather than an ansatz limitation — the circuit is capable of representing the correct state, but cold-start gradient descent has no mechanism to switch branches. The entanglement analysis was clean up to the crossing: the von Neumann entropy correctly distinguishes product states from maximally entangled Bell states, and in the two-qubit Hamiltonian it rises sharply near the level repulsion, where the ground state becomes a strongly correlated superposition. This connection between level repulsion and entanglement growth is one of the cleaner results in the project.

The Lipkin model is where things get harder. Classical diagonalization of both the $J = 1$ and $J = 2$ Hamiltonians works without difficulty, and the Hartree-Fock comparison confirms the expected behavior: exact below the critical coupling, systematically biased above it. The VQE also handles $J = 1$ well. For $J = 2$ the residual error of $\approx 10^{-2}$ reflects a genuine structural limitation: the four-qubit Hilbert space contains three symmetry sectors, and the hardware-efficient ansatz has no mechanism to stay within the physical one. The problem is worst near the phase transition, where the optimization landscape is locally flat and gradient-based methods stall. Four random restarts help but do not fix it. These results are also obtained under ideal conditions — exact statevector simulation, no gate noise. On real hardware the precision would be worse, likely by a significant margin at circuit depth 4.

5.2 Reflections

Working through this progression made the formalism more concrete in ways that are hard to get from reading alone. The link between entanglement and interaction strength — visible in the entropy curves as λ passes through the crossing region — and the way the Pauli decomposition translates a physical Hamiltonian into a form that can be measured circuit-by-circuit both became much clearer through the implementation. The VQE errors were instructive in a way I did not fully anticipate. For the two-level system, cold-start BFGS converges immediately and perfectly. For the two-qubit system, the optimizer succeeds up to the avoided crossing, then fails in a precisely located and physically interpretable way — the crossing point where the ground state switches character. For the Lipkin $J = 2$ case the failure is again tied to the physics, this time the near-flat landscape near the phase transition rather than a branch discontinuity. In all three cases the optimization difficulty maps directly onto the structure of the problem, which I found more interesting than a generic convergence failure would have been.

Having two independent VQE implementations — manual and Qiskit-based — agree throughout was also reassuring: to machine precision for $\lambda \lesssim 0.4$ in the two-qubit case, and identically in failure for $\lambda > 0.4$. The latter is particularly useful — both implementations making the same mistake at the same crossing point confirms the behavior is physical, not a software artifact, and ruled out implementation errors before moving to the more demanding Lipkin system.

5.3 Future Work

For the two-qubit system, the branch-tracking failure at $\lambda \approx 0.4$ could be addressed by warm-starting each λ step from the previous solution and supplementing with random restarts near the crossing, or by using an excited-state VQE variant that explicitly targets the lower branch. For the $J = 2$ Lipkin system, the most direct improvement would be a symmetry-adapted VQE using a circuit constrained to the $J = 2$ sector from the start. This would likely close the $\approx 10^{-2}$ precision gap and scale more favorably to larger N. A natural extension from there is a sweep over both V and W, mapping the full two-parameter phase diagram of the LMG model — the current implementation supports arbitrary W, so this is mostly a matter of running the experiments. Finally, testing these circuits on actual quantum hardware via the IBM Quantum network would expose the gap between statevector simulation and real device performance, and would make the limitations identified here more concrete.

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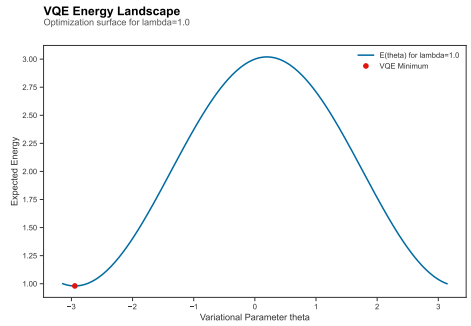
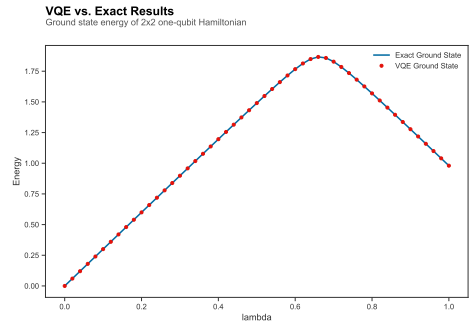
6 Appendix

The figures in this appendix supplement those in the main text. Figures already shown in the Results section are not repeated here.

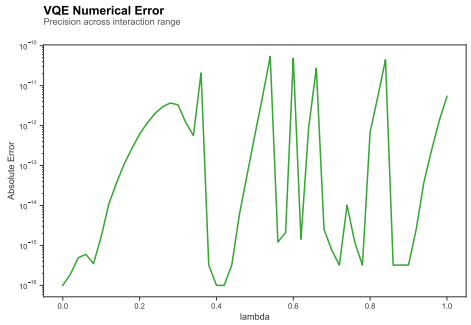
Part C: Additional VQE diagnostics for the two-level model



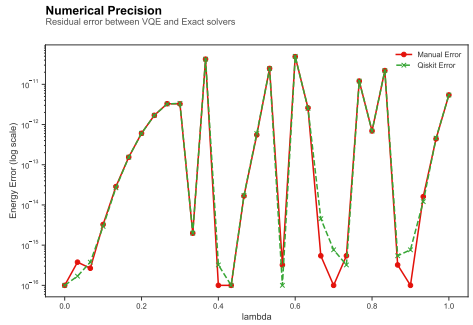
(a) Energy landscape $E(\theta)$ for the R_y ansatz at selected values of λ, showing the smooth, unimodal structure that ensures reliable BFGS convergence.



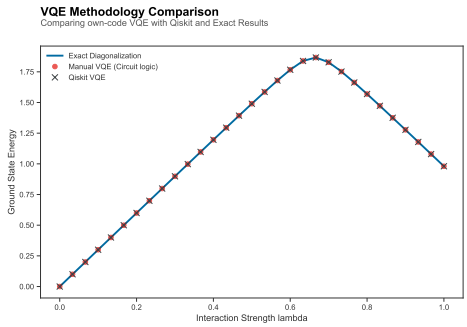
(b) VQE ground state energy vs. exact diagonalization across $\lambda \in [0, 1]$.



(c) Absolute VQE error as a function of λ for the single-qubit model.



(d) Log-scale precision plot showing error remaining below 10^{-10} throughout.



(e) Comparison between manual VQE implementation and Qiskit `SparsePauliOp` for the two-level model.

Figure 19. Supplementary VQE diagnostics for the 2×2 two-level Hamiltonian (Part c).

Part E: Additional VQE diagnostics for the two-qubit model

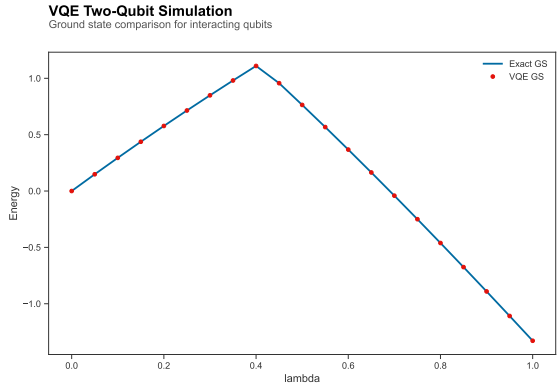
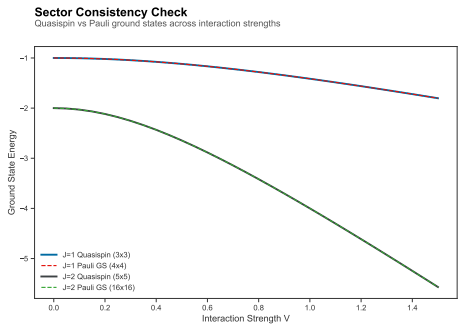
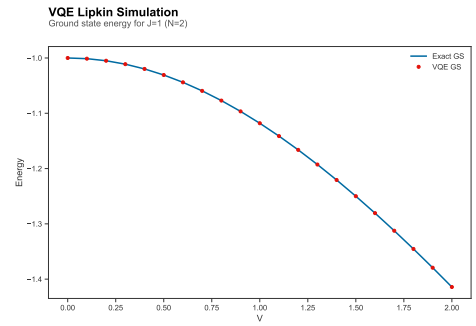


Figure 20. VQE ground state energy vs. exact diagonalization for the two-qubit Hamiltonian, showing agreement across the full λ range.

Part G: Additional Lipkin VQE diagnostics



(a) Verification that the VQE correctly identifies eigenvalues within the $J = 2$ sector of the full 16×16 qubit Hilbert space, matching the quasispin reference values.



(b) VQE ground state energy for the Lipkin $J = 1$ system as a function of V, with residual error at the 10^{-11} level.

Figure 21. Supplementary diagnostics for the Lipkin VQE (Part g).

6.1 Circuits

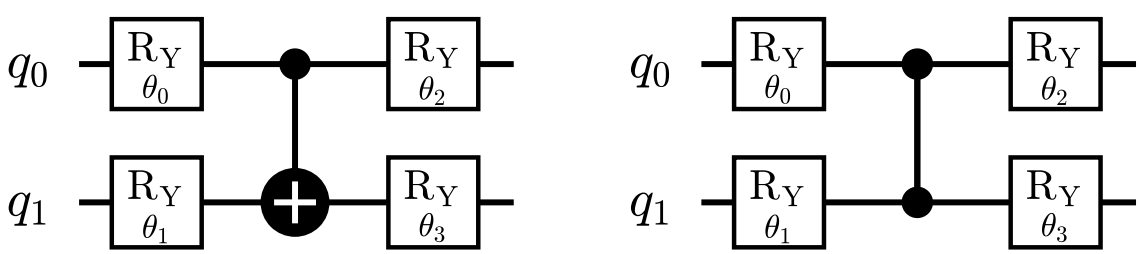


Figure 22. Quantum circuits for all configurations

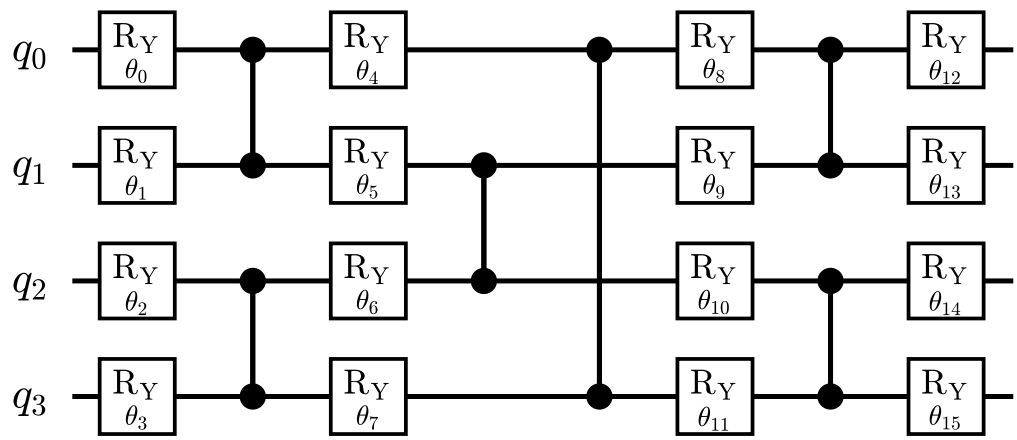


Figure 23. Quantum circuits for all configurations

Python environment

```
matplotlib>=3.7.1
numpy>=1.24.3
pandas>=2.2.2
pennylane>=0.38.0
qiskit>=1.0.0
qiskit-aer>=0.17.2
qiskit-algorithms>=0.4.0
scipy>=1.14.1
seaborn>=0.13.2
sympy>=1.13.3
```

Listing 2. Python environment and core libraries used for the numerical simulations, classical diagonalization, and quantum-circuit implementations in this project.

VQE Algorithm

Algorithm 1 gives a self-contained description of the VQE procedure as implemented in this project. All source code is available in the project repository (Furnes 2026b; Furnes 2026a).

Algorithm 1 Variational Quantum Eigensolver (VQE)

Require: Hamiltonian $\mathcal{H} = \sum_k c_k P_k$, ansatz $U(\boldsymbol{\theta})$, initial parameters $\boldsymbol{\theta}_0$
Ensure: Approximate ground state energy E_{VQE}

```
Decompose H into Pauli strings {P_k} with real coefficients {c_k}
Initialize θ ← θ_0
repeat
    Prepare trial state |ψ(θ)⟩ = U(θ) |0⟩^⊗n
    for each Pauli term P_k do
        Rotate to the appropriate measurement basis
        Evaluate expectation value ⟨P_k⟩_θ
    end for
    Compute energy E(θ) = Σ_k c_k ⟨P_k⟩_θ
    Update θ via classical optimizer (BFGS / L-BFGS-B)
until |E(θ_{t+1}) − E(θ_t)| < ε_tol
return E_VQE = E(θ*)
```